Data structures

Programming · Level 4 · Lesson 3



In this lesson you will

- Choose a list, stack, queue, set or dictionary by the operations a job needs.
- Use a stack to check brackets.
- Count with a dictionary.

Key ideas

- Stack: last in, first out. Queue: first in, first out.
- x in a list: O(n). x in a set: about one step.
- count[w] = count.get(w, 0) + 1

Activity: Bracket stack

1. Warm-up (3 min): A pile of books: push and pop.
2. Check (7 min): Use paper slips as a stack to check brackets.
3. Break (7 min): Write bracket strings that break a naive checker.
4. Code (3 min): Run your strings through balanced().

Paper slips make a perfect stack: only the top one can be taken.

Let's discuss

1. Why is checking a set so much faster than checking a list?
2. When would a queue be the wrong choice?
3. What does a set give up in exchange for its speed?

Exit ticket

- Name the structure for each: a browser’s back button, a printer’s jobs, a word count.